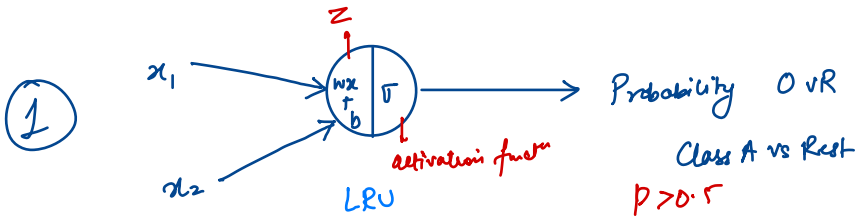
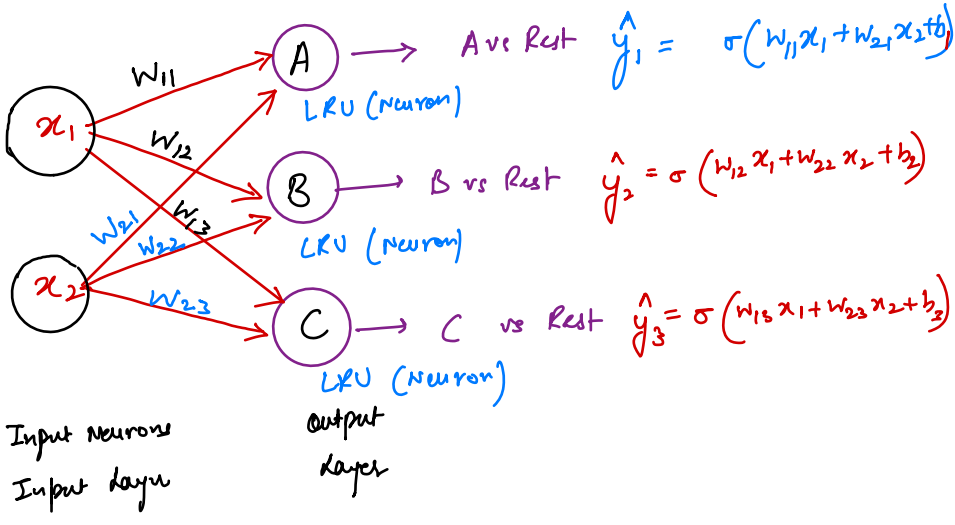
Agenda

- Model development
- Classifier
- Multi - Class
- Traing your first Model
  - forward & Backword prob.

Can we use simple logistic regression to classify 3 classes?





$$\hat{y} = \sigma(z) \quad z = Wx + b$$

σ [0, 1]

Ideally, $\hat{y}_1 + \hat{y}_2 + \hat{y}_3 = 1$

Sigmoid → Binary class classification

Class A vs Rest

1000

$p > 0.6$ (A)

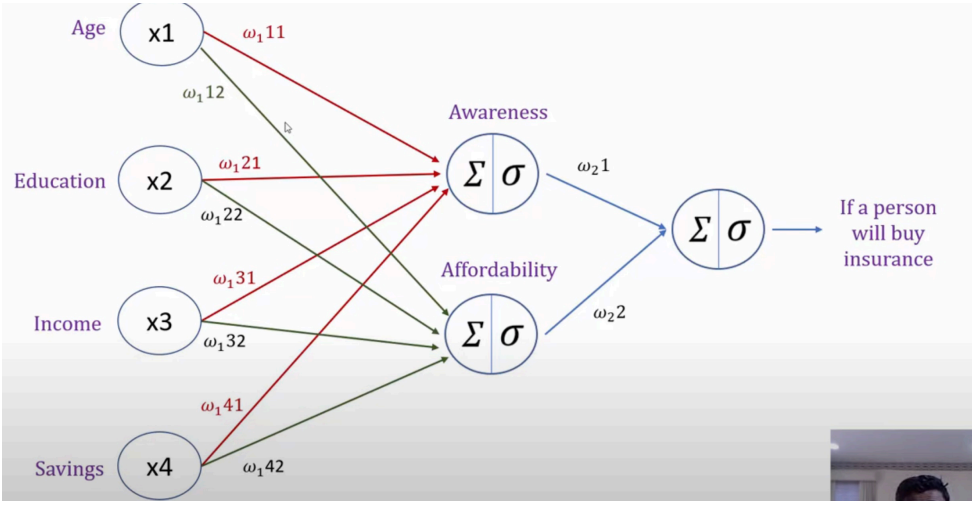
0.4

Class B vs Rest

$p > 0.5$

0.5

Multi-class classification → Softmax function



Let's say 3 class Classification

Softmax => $P_A = \frac{e^{z_A}}{e^{z_A} + e^{z_B} + e^{z_C}}$

$P_B = \frac{e^{z_B}}{e^{z_A} + e^{z_B} + e^{z_C}}$

$P_C =$

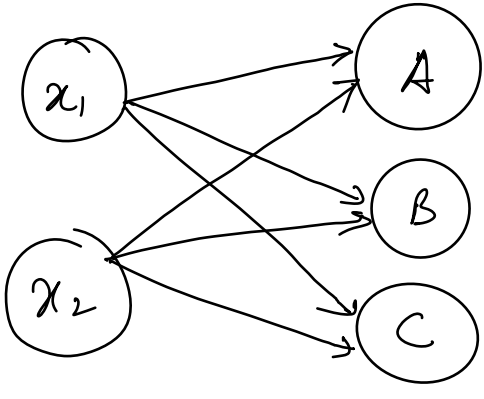
Imagine → This is Binary

A vs B

$P_A = \frac{e^{z_A}}{e^{z_A} + e^{z_B}} = \frac{e^{z_A}(1)}{e^{z_A}\left(1 + \frac{e^{z_B}}{e^{z_A}}\right)}$

$$P_A = \frac{1}{1 + e^{(z_B - z_A) = z_{A'}}}$$

$$\rightarrow P_A = \frac{1}{1 + e^{z_{A'}}} \quad (-z)$$



$$z = \begin{bmatrix} 2 \\ 6 \\ 12 \end{bmatrix} \rightarrow P = \begin{bmatrix} 0.0004 \\ 0.006 \\ 0.99.. \end{bmatrix} \quad \hat{y} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Softmax

→ (d) (shape of x) = 2

(n) (no. of class) = 3

(m) (no. of rows) = 300

→ Size of weight Matrix ✓

$$d \times n$$

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}$$

(2×3)

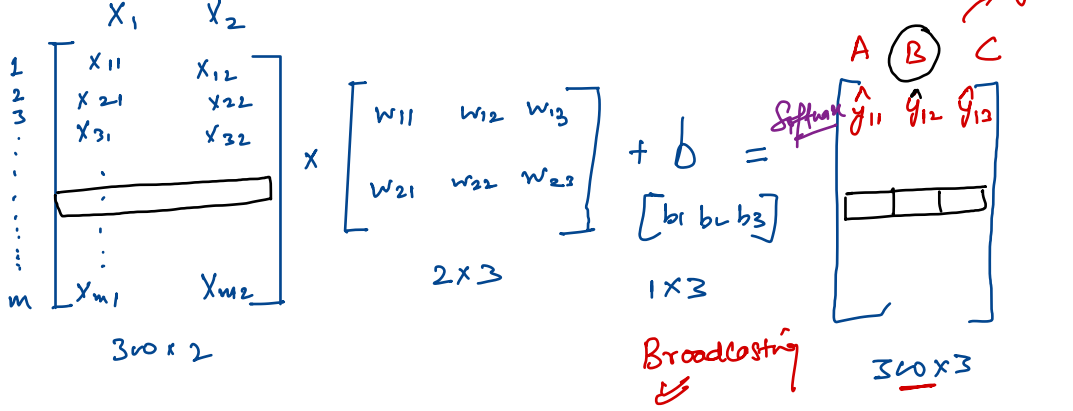
→ Size of input ⇒ Rows × Colum

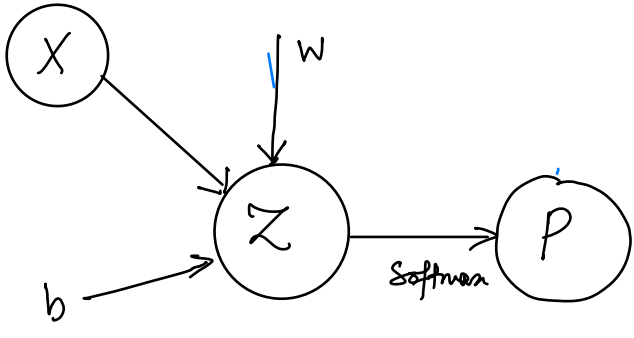
300 × (x_1, x_2)

$m = 300$

⇒ 300 × 2

⇒ m × d





Loss

$L =$ Actuals with Prob

$y \log p$ ← Categorical Cross Entropy

$y \log p + (1-y) \log (1-p)$

$W_{new} = W_{old} - \frac{\partial L}{\partial w}$

If Binary

$$L = y \log p + (1-y) \log(1-p)$$

$$= y \log \sigma(z)$$

$$= y \log \left(\frac{1}{1+e^{-z}} \right)$$

$$= y \log \frac{1}{1+e^{-(wx+b)}}$$

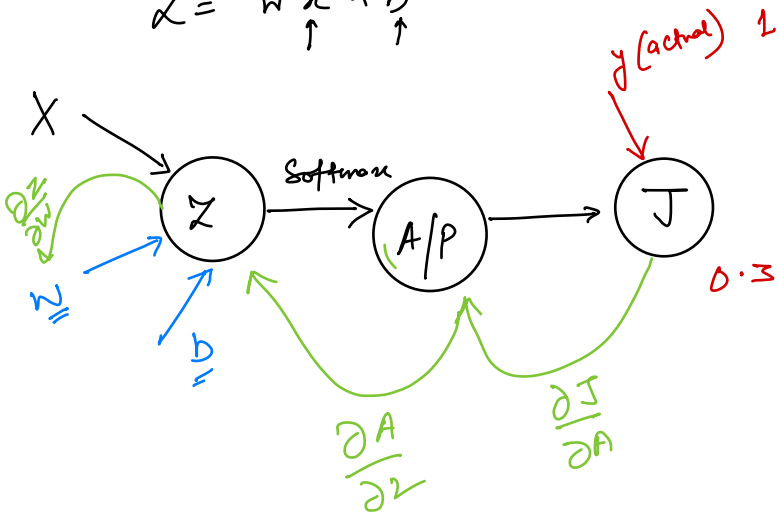
$$\frac{\partial L}{\partial w} =$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial p} \times \frac{\partial p}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\text{Loss} = -\sum_{i=1}^{\text{output size}} y_i \cdot \log \hat{y}_i$$

Multiclass Cross-Entropy

$$Z = wx + b$$



$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial A} \times \frac{\partial A}{\partial Z} \times \frac{\partial Z}{\partial w}$$

Backward prop

$Z = wX + b$

$A = \text{Softmax}(Z)$

$\mathcal{L} = y \log A$

Forward Prop

